A G^R_D ON A PROPER CURVE WITH H^0(O_X) = K SATISFIES R <= D, WITH EQUALITY FORCING GENUS 0

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the section-sequence dimension count with the triviality of L|_D, the equality-case strategy remark, the doubling step with its explicit dimension identity, and the closing ampleness argument are all retained verbatim, including the author's footnote on the triviality of L|_D.

Editorial note. The label lemma-grd-inequalities names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Cartier divisors from sections. Divisors, Lemma divisors-lemma-characterize-OD and Remark divisors-remark-ses-regular-section; Varieties, Lemma varieties-lemma-degree-effective-Cartier-divisor. Supplies the sequence 0 -> O_X -> L -> L|_D -> 0 for the zero divisor D of a section, with deg(D) = deg(L) = d.
- Ampleness and vanishing on curves. Varieties, Lemma varieties-lemma-ample-curve; Cohomology of Schemes, Lemma coherent-lemma-coherent-proper-ample. Provides the power L^tensor n with vanishing H^1 used to close the equality case.

Source reference. Original lemma 0B5T: sources/stacks-chow/tags.txt, source lines 14046–14046. Prerequisite pointer invoked at proof line(s) 731; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0B9W: sources/stacks-curves/pinned-original-source.tex, source lines 1677–1677. Prerequisite pointer invoked at proof line(s) 664; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01X0: sources/stacks-chow/tags.txt, source lines 2501–2501. Prerequisite pointer invoked at proof line(s) 676; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BUT: sources/stacks-chow/tags.txt, source lines 14886–14886. Prerequisite pointer invoked at proof line(s) 683; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0C6K: sources/stacks-chow/tags.txt, source lines 15300–15300. Prerequisite pointer invoked at proof line(s) 677; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0B5X: sources/stacks-chow/tags.txt, source lines 14050–14050. Prerequisite pointer invoked at proof line(s) 728; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AYY: sources/stacks-chow/tags.txt, source lines 13806–13806. Prerequisite pointer invoked at proof line(s) 678; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper curve over k with $H^0(X, \mathcal{O}_X) = k$. If X has a $\mathfrak{g}^r_d$, then $r \leq d$. If equality holds, then $H^1(X, \mathcal{O}_X) = 0$, i.e., the genus of X (Definition 0B9W) is* 0.

Proof. Let $(\mathcal{L}, V)$ be a $\mathfrak{g}^r_d$. Since this will only increase r, we may assume $V = H^0(X, \mathcal{L})$. Choose a nonzero element $s \in V$. Then the zero scheme of s is an effective Cartier divisor $D \subset X$, we have $\mathcal{L} = \mathcal{O}_X(D)$, and we have a short exact sequence

$$0 \to \mathcal{O}_X \to \mathcal{L} \to \mathcal{L}|_D \to 0$$

see Divisors, Lemma 01X0 and Remark 0C6K. By Varieties, Lemma 0AYY we have $\deg(D) = \deg(\mathcal{L}) = d$. Since D is an Artinian scheme we have $\mathcal{L}|_D \cong \mathcal{O}_D$[1]. Thus

$$\dim_k H^0(D, \mathcal{L}|_D) = \dim_k H^0(D, \mathcal{O}_D) = \deg(D) = d$$

On the other hand, by assumption $\dim_k H^0(X, \mathcal{O}_X) = 1$ and $\dim H^0(X, \mathcal{L}) = r+1$. We conclude that $r + 1 \leq 1 + d$, i.e., $r \leq d$ as in the lemma.

Assume equality holds. Then $H^0(X, \mathcal{L}) \to H^0(X, \mathcal{L}|_D)$ is surjective. If we knew that $H^1(X, \mathcal{L})$ was zero, then we would conclude that $H^1(X, \mathcal{O}_X)$ is zero by the long exact cohomology sequence and the proof would be complete. Our strategy will be to replace $\mathcal{L}$ by a large power which has vanishing. As $\mathcal{L}|_D$ is the trivial invertible module (see above), we can find a section t of $\mathcal{L}$ whose restriction of D generates $\mathcal{L}|_D$. Consider the multiplication map

$$\mu : H^0(X, \mathcal{L}) \otimes_k H^0(X, \mathcal{L}) \longrightarrow H^0(X, \mathcal{L}^{\otimes 2})$$

and consider the short exact sequence

$$0 \to \mathcal{L} \xrightarrow{s} \mathcal{L}^{\otimes 2} \to \mathcal{L}^{\otimes 2}|_D \to 0$$

Since $H^0(\mathcal{L}) \to H^0(\mathcal{L}|_D)$ is surjective and since t maps to a trivialization of $\mathcal{L}|_D$ we see that $\mu(H^0(X, \mathcal{L}) \otimes t)$ gives a subspace of $H^0(X, \mathcal{L}^{\otimes 2})$ surjecting onto the global sections of $\mathcal{L}^{\otimes 2}|_D$. Thus we see that

$$\dim H^0(X, \mathcal{L}^{\otimes 2}) = r + 1 + d = 2r + 1 = \deg(\mathcal{L}^{\otimes 2}) + 1$$

Ok, so $\mathcal{L}^{\otimes 2}$ has the same property as $\mathcal{L}$, i.e., that the dimension of the space of global sections is equal to the degree plus one. Since $\mathcal{L}$ is ample (Varieties, Lemma 0B5X) there exists some n_0 such that $\mathcal{L}^{\otimes n}$ has vanishing H^1 for all $n \geq n_0$ (Cohomology of Schemes, Lemma 0B5T). Thus applying the argument above to $\mathcal{L}^{\otimes n}$ with $n = 2^m$ for some sufficiently large m we conclude the lemma is true. □

[1] In our case this follows from Divisors, Lemma 0BUT as $D \to \mathrm{Spec}(k)$ is finite.